

Imitation Learning Model

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1 Input/Output Structure

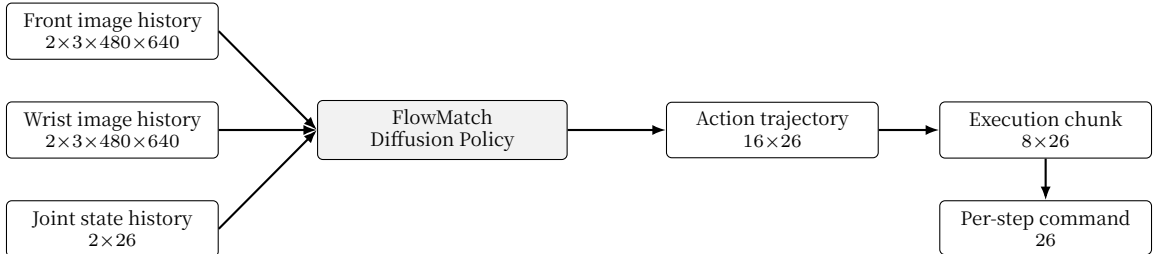
This policy is a visuomotor imitation learning model trained from demonstration trajectories. The model conditions on a short observation history and generates a short-horizon action trajectory. In the current configuration, $n_{\text{obs}} = 2$, $H = 16$, and $n_{\text{act}} = 8$.

$$o_{t-1:t} = \left(I_{t-1:t}^{\text{front}}, I_{t-1:t}^{\text{wrist}}, s_{t-1:t} \right), \quad \hat{A}_{t:t+H-1} = \pi_{\theta}(o_{t-1:t})$$

Feature	Temporal Shape	Role
Front RGB image history	$2 \times 3 \times 480 \times 640$	Global visual context
Wrist RGB image history	$2 \times 3 \times 480 \times 640$	Local visual context
Robot state history	2×26	Arm 6 DoF + hand 20 DoF state
Generated action trajectory	16×26	Horizon-level action prediction
Execution action chunk	8×26	Actions queued for execution
Per-step command	26	One arm/hand joint target sent at a control step

Table 1: Temporal input/output structure

The state and action share the same 26D joint space. The model does not output a Cartesian waypoint or a symbolic command. It predicts target joint vectors directly, with arm motion and hand motion represented as one coordinated action space.



2 Temporal Action Generation

The generated trajectory has shape $H \times D_a = 16 \times 26$. At inference time, the policy extracts the execution window starting from the current observation index:

$$\hat{A}_{t:t+15} \in \mathbb{R}^{16 \times 26}, \quad A_{t:t+7}^{\text{exec}} \in \mathbb{R}^{8 \times 26}, \quad a_t \in \mathbb{R}^{26}.$$

Thus, the model-level output is a short action trajectory, while the control-level output at each step is a single 26D joint command. This design preserves short-term motion intent and temporal consistency across successive arm/hand commands.